PARTHASARATHY GHOSE

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Former Professor and Academic Programme Coordinator S. N. Bose National Centre for Basic Sciences

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Profile

Parthasarathy Ghose is a theoretical physicist, historian and philosopher of science whose research career spans more than six decades. Trained under National Professor Satyendra Nath Bose, he has made contributions to particle physics, quantum field theory, foundations of quantum mechanics, Bohmian mechanics, stochastic approaches to quantum theory, classical entanglement, quantum cognition, neural dynamics, and philosophy of science.

His work has appeared in journals including *Physical Review*, *Physical Review A*, *Physical Review D*, *Physics Letters A*, *Physics Letters B*, *Foundations of Physics*, *Pramana*, *Physica A*, *Computational and Structural Biotechnology Journal*, and *Physica Scripta*.

He is the author or editor of numerous books, including *Testing Quantum Mechanics on New Ground*, *Einstein, Tagore and the Nature of Reality*, and *An Interdisciplinary Approach to Cognitive Modelling*.

Research Profile

- Research career spanning more than six decades.
- More than 150 research publications, books, and scholarly articles.
- Over 1,000 citations.
- h-index: 18.
- Research contributions spanning particle physics, quantum foundations, classical entanglement, quantum cognition, neural dynamics, and philosophy of science.

Education

- Ph.D. (Science), University of Calcutta (1968) Supervisor: Professor S. N. Bose, FRS
- B.Sc. (Special), Imperial College London (1963)
- B.Sc. (Honours), Presidency College, University of Calcutta (1961)

Academic Positions

1963–1968 Junior Scientist under National Professor S. N. Bose

1968–1975 Lecturer and Reader, Visva-Bharati University

1975–1989 Science Officer, British Council, Kolkata

1989–1999	Professor and Academic Programme Coordinator, S. N. Bose National Centre for Basic Sciences
1999–2003	Honorary Consultant (Academic), S. N. Bose National Centre for Basic Sciences
2010–2019	Senior Scientist (Platinum Jubilee Fellow), National Academy of Sciences, India
Present	Distinguished Fellow, Tagore Centre for Natural Sciences and Philosophy

Research Areas

- Foundations of Quantum Mechanics
- Quantum Field Theory
- Bohmian Mechanics
- Nelson Stochastic Mechanics
- Quantum–Classical Transition
- Classical Entanglement and Contextuality
- Quantum Cognition
- Neural Dynamics and Quantum-like Processes
- Philosophy of Science
- Science and Consciousness
- History and Philosophy of Modern Physics

Honours and Distinctions

- Honorary Member of the Most Excellent Order of the British Empire (MBE), 1990
- Fellow, National Academy of Sciences, India
- Fellow, West Bengal Academy of Science and Technology
- Indira Gandhi Prize for Popularization of Science
- National Award for Science Communication
- Ramendra Sundar Trivedi Puraskar
- Vice-President, Indian Physical Society
- Chairman, Indian Physics Association (Calcutta Chapter)

Selected Research Contributions

Particle Physics and Quantum Field Theory

Research on broken SU(3) symmetry, vector meson dominance, gauge theories, grand unification, Higgs mass corrections, QCD sum rules, quark confinement, and quark–gluon plasma physics.

Foundations of Quantum Mechanics

Contributions to wave–particle duality, complementarity, EPR correlations, Bohmian mechanics, relativistic quantum mechanics of bosons, spontaneous collapse, and quantum–classical transitions.

Classical Entanglement and Contextuality

Among the early contributors to Hilbert-space formulations of classical optics and studies of entanglement-like phenomena in classical electromagnetic fields.

Quantum Cognition

Development of interdisciplinary frameworks connecting cognition, probability theory, philosophy, and quantum-inspired decision models.

Neural Dynamics and Quantum-like Models

Recent work explores stochastic and quantum-like markers in neural dynamics, including quantum noise in neuronal models and temporal quantum correlations.

Cosmology

Recent investigations of Hamiltonian monodromy and quantum spectra in cosmological minisuperspace models.

Selected Recent Publications (2016–2026)

1. A. K. Rajagopal and P. Ghose, "Hilbert Space Theory of Classical Electrodynamics," *Pramana* **86**, 1161–1172 (2016).

2. A. Banerjee et al. and P. Ghose, "Study on Brain Dynamics by Nonlinear Analysis of Music-Induced EEG Signals," *Physica A* **444**, 110–120 (2016).

3. S. Patra and P. Ghose, "Quantum-like Modelling in Game Theory: Quo Vadis? A Brief Review," *Asian Journal of Economics and Banking* **4**(3), 49–66 (2020). doi: 10.1108/AJEB-08-2020-0054.

4. S. Patra and P. Ghose, "Measurement, Lüders and von Neumann Projections and Nonlocality," *Pramana – Journal of Physics* **96**, 34 (2022).

5. P. Ghose and S. Patra, *An Interdisciplinary Approach to Cognitive Modelling: A Framework Based on Philosophy and Modern Science*, Routledge (2023).

6. P. Ghose and Y. Mirgorod, "Relationship between Classical and Quantum Mechanics in Micellar Aqueous Solutions of Surfactants," *Physics of Wave Phenomena* **32**, 34–42 (2024).

7. P. Ghose and D. A. Pinotsis, "The FitzHugh–Nagumo Equations and Quantum Noise," *Computational and Structural Biotechnology Journal* **30**, 12–20 (2025).

8. P. Ghose, "Bose's Probabilistic Interactions, Einstein's Objections, and Quantum Optics Legacy," *Academia Quantum* **2**(4) (2025).

9. I. Ghorai, H. Soni, A. Chakraborty and P. Ghose, "Quantum Monodromy in a Cosmological Minisuperspace," *Physica Scripta* **101**, ae7039 (2026).

Books

Research Monographs

- *Testing Quantum Mechanics on New Ground*, Cambridge University Press (1999)
- *An Interdisciplinary Approach to Cognitive Modelling*, Routledge (2023)

Edited Volumes

- *Einstein, Tagore and the Nature of Reality*, Routledge (2016)
- *Tagore, Einstein and the Nature of Reality: Literary and Philosophical Reflections*, Routledge (2019)

Popular Science and Literature

- *Riddles in Your Tea Cup*
- *Cosmic Quest*
- *Kanamatra* (2025)
- *Bhuture Beral* (2026)

Science Communication

Author of numerous books and articles on modern physics, cosmology, quantum theory, and the history and philosophy of science. Recipient of national awards for science communication and public understanding of science.

Current Research Programme

- Nelson Stochastic Mechanics
- Quantum–Classical Interpolation
- Bell Inequalities and Contextuality
- Classical Entanglement
- Quantum Cognition
- Process-Theoretic Models of Cognition
- Neural Leggett–Garg Tests
- Persistent Stochastic Electrodynamics
- Foundations of Measurement
- Mind–Brain–Consciousness Studies

Recent Preprints and Working Papers (2024–2026)

The following recent preprints reflect ongoing research in quantum foundations, logic, contextuality, cognition, neuroscience, and stochastic approaches to physical theory.

1. P. Ghose, "Intersystem Non-separability and CHSH-Bell Violations in Classical Optics," arXiv (2024).
2. P. Ghose and S. Patra, "Context and Complementarity: Formalizing Bohr's Vision through Many-Valued Contextual Logic," arXiv (2025).
3. P. Ghose, "A Contextual Seven-Valued Logic (Saptabhanginaya) for Quantum Systems," arXiv (2025).
4. P. Ghose, "Einstein's Worries and Actual Physics: Beyond Pilot Waves," arXiv (2026).
5. P. Ghose, "Indefinite Causal Order from Failure-to-Glue: Contextual Semantics and Parametric Time," arXiv (2026).
6. P. Ghose, "Rethinking Nonlocality: Locality, Counterfactuals, and the EPR–Bell Argument," arXiv (2026).
7. P. Ghose, "Nelson's Stochastic Mechanics: Measurement, Nonlocality, and the Classical Limit," arXiv (2026).
8. P. Ghose, "Leggett–Garg Tests in Neural Dynamics: Probing Non-Diffusive Stochastic Structure in Single Neurons," arXiv (2026).
9. P. Ghose, "Bell-like States in Classical Optics: A Process-Theoretic and Sheaf-Theoretic (Categorical) Clarification," arXiv (2026).
10. P. Ghose, "Measurement as Sheafification: Context, Logic, and Truth after Quantum Mechanics," arXiv (2026).

Recent Academic Activities at IIT Mandi (2025–Present)

In recent years Professor Ghose has been actively involved in interdisciplinary programmes at IIT Mandi on quantum cognition, neural dynamics, and mind–brain–consciousness studies, bringing together physics, neuroscience, psychology, and philosophy.

- Developed and taught undergraduate courses on the History and Philosophy of Science and Technology for first-year B.Tech. students.
- Designed and delivered a fourteen-lecture postgraduate course on Quantum Cognition for doctoral students and faculty from multiple disciplines, including psychology, cognitive science, physics, and philosophy.
- Contributed to the conceptual development of an interdisciplinary research programme on Mind–Brain–Consciousness studies, integrating experimental neuroscience, cognitive science, philosophy of mind, and theoretical physics.
- Delivered invited lectures and keynote presentations on quantum foundations, cognition, neural dynamics, and consciousness studies.

- Initiated collaborative research on stochastic-process models of neural dynamics, quantum-like markers in neuronal systems, and finite-speed signal propagation in biological networks.
- Participated in the Mind–Brain–Consciousness Conference, IIT Mandi (June 2026), delivering a Keynote Address and contributing to discussions on neuroscience, cognition, and foundational questions concerning consciousness.
- Mentored students and young researchers in quantum foundations, quantum cognition, neuroscience, and philosophy of science.